A Reciprocal-Shell Zero-Sum Contrast: Finite Scale Repair for Prime-Incidence Defect Witnesses

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Abstract

TPC-350 found that a fixed balanced-step contrast of prime-divisibility incidences has a positive finite response but a low floor on large prime shells. We test one predeclared replacement on the identical literal masked operator and the identical 192-row panel. For shell primes $p_0 < \cdots < p_{r-1}$, set $\gamma_j = p_j^{-1} - r^{-1}\sum_k p_k^{-1}$. This rational rule is independent of the interval, source law, and observed matrix, and has exact zero sum. Its prime-incidence Gram identity and induced-norm lower witness are exact. All 192 reciprocal witnesses have positive response; 180 improve the locked parent response, 111 reach half the defect norm, and 86 beat the coordinate baseline. The response-to-defect ratio ranges from 0.0917557319271 to 0.901734353382, with mean 0.539037202287. At $Q = 256$, four rows now reach one half, compared with none for the parent, but the minimum remains below one quarter. Thus the reciprocal rule gives a genuine finite scale repair, not a uniform floor, arithmetic L^2 estimate, or twin-prime result.

1 Question and claim boundary

The balanced-step witness in TPC-350 survived three fresh origins and four finite interval lengths, yet its $Q = 256$ block had no row reaching one half of the defect norm and a minimum ratio 0.0657381187306. The smallest natural continuation is to change only the shell coefficient rule while retaining the literal masked object and every panel key.

We impose a strict no-fit condition: coefficients may depend on the shell primes, but not on the origin, interval length, source signs, kernel exponent, matrix entries, singular vectors, or response. All comparisons below are finite. They do not imply a limit as M or Q grows, a source-uniform arithmetic estimate, a fixed power of x, or a statement about twin primes.

2 Literal masked defect

Let $I = \{o, o+1, \ldots, o+M-1\}$ and let R_I, E_I denote restriction and zero extension. For a shell prime p, put $(P_p f)(n) = \mathbf{1}_{p\nmid n} f(n)$. The physical matrix, its unmasked ideal, and their defect are

$$A_I = \sum_p \varepsilon_p R_I P_p K_p P_p E_I,$$

$$T_I = R_I \left(\sum_p \varepsilon_p K_p \right) E_I, \qquad D_I = A_I - T_I.$$

The source signs ε_p are fixed independently of the test coefficients. The finite matrix entry is

$$\mathbf{1}_{u\neq t}\mathbf{1}_{p\nmid u}\mathbf{1}_{p\nmid t}\, p h_s(u-t) \left(\mathbf{1}_{p|u-t} - \frac{1}{p-1} \right), \qquad h_s(d) = \frac{H^{2s}}{(H^2+d^2)^s}, \quad H = 66.$$

3 Reciprocal-shell incidence interface

For the primes in $(Q, 2Q]$, define

$$\gamma_j = \frac{1}{p_j} - \frac{1}{r}\sum_{k=0}^{r-1}\frac{1}{p_k}. \tag{1}$$

For $t \in I$, let $h_{p_j,I}(t) = \mathbf{1}_{p_j|t}$ and set

$$c_I = \sum_{j=0}^{r-1}\gamma_j h_{p_j,I}.$$

Multiply-divisible positions retain the algebraic sum of all incidences.

Proposition 1 (exact balance)**.** *The rational coefficient vector in* (1) *satisfies* $\sum_j \gamma_j = 0$*.*

Proof. The first terms sum to $\sum_j p_j^{-1}$, while the repeated mean contributes $r \cdot r^{-1}\sum_k p_k^{-1}$. ☐

Proposition 2 (reciprocal incidence Gram identity)**.** *For every finite matrix* D_I*,*

$$\|D_I c_I\|_2^2 = \sum_{j,k}\gamma_j\gamma_k\langle D_I h_{p_j,I}, D_I h_{p_k,I}\rangle. \tag{2}$$

Proof. Apply linearity to the finite sum defining c_I and expand the Euclidean inner product. ☐

Theorem 1 (normalized lower witness)**.** *If* $c_I \neq 0$*, then*

$$\|D_I\|_{2\to 2} \geq \frac{\|D_I c_I\|_2}{\|c_I\|_2}. \tag{3}$$

Proof. The vector $c_I/\|c_I\|_2$ is a unit vector. Insert it into the definition of the induced Euclidean norm; see also [1]. ☐

Remark 1. *Equations* (1)–(3) *are exact finite algebra. They do not assert a sign for the cross-prime Gram terms or cancellation in the source variable.*

4 Frozen paired protocol

The TPC-350 producer and certificate are normalized-hash locked. We retain

$$o \in \{60097, 72097, 84097\}, \quad M \in \{256, 512, 1024, 2048\}, \quad Q \in \{36, 80, 128, 256\},$$

the exponents $s \in \{1, 2\}$ and source laws `all_plus` and `alternating_index`. Hence there are 192 paired rows and 48 fixed (o, Q, s, law) length series. Each reciprocal response is compared with the parent balanced-step response on exactly the same row key. The coordinate control is the largest defect column norm among mask-hit positions.

The producer accumulates shells forward. An independent checker accumulates them in reverse, recomputes all matrices, exact coefficients, norm ratios, parent comparisons, length series, and a rational anchor. No producer module is imported by the independent checker.

5 Finite results

All reciprocal witnesses are nonzero with positive response. The parent comparison is broad but not universal: 180 rows improve and 12 do not. The half-defect census rises from 91 to 111, while the coordinate-baseline census rises from 70 to 86. Grouping by shell scale gives the sharper readout:

Q	reciprocal min	reciprocal mean	reciprocal max	half	improved
36	0.688158407450	0.804654885138	0.901734353382	48	48
80	0.364933284503	0.632666956348	0.834679518677	41	48
128	0.178852728310	0.448071439797	0.783459763859	18	44
256	0.0917557319271	0.270755527864	0.537734585425	4	40

Table 1: TPC-351 paired finite audit.

Quantity	Certified finite readout
Rows / length series	192 / 48
Positive reciprocal responses	192/192
Rows improving TPC-350	180/192
Reciprocal incidence support	24–339
$\|D_I x_I\|/\|D_I\|$	0.0917557319271–0.901734353382
Mean response-to-defect ratio	0.539037202287
Coordinate baseline beaten	86/192
At least half the defect norm	111/192
Nondecreasing length series	25/48
Arithmetic advance / fixed-power credit	no / zero

The $Q = 256$ block is a real finite repair: its minimum increases from 0.0657381187306 to 0.0917557319271, and four rows reach one half instead of zero. It is not a universal repair, because eight high-shell rows lose to the parent and the block still contains sub-quarter values. Only 25 of 48 length series are nondecreasing, one more than the parent but far from a monotonicity theorem.

5.1 Exact reciprocal anchor

For $I = [97, 110]$, $Q = 4$, $s = 1$, and all-plus source signs, the shell is $(5, 7)$ and $\gamma = (1/35, -1/35)$. The exact incidence vector is

$$c_I = \frac{1}{35}(0, -1, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1), \qquad \|c_I\|_2^2 = \frac{3}{1225}.$$

Its exact squared response is

$$\|D_I c_I\|_2^2 = \frac{1427659395645308157177240916237167455767195256668781964853384 4297}{15411127350108125013094996416827239513110623503288437540341250 0000}.$$

This anchor checks rational centering, multiply-divisible incidences, and the literal defect without floating point.

6 Adversarial validation and claim boundary

Normal and optimized producer, independent, stress, and Bridge-B outputs must match byte for byte. Mutation tests reject a changed panel, deleted high shell, false positive census, false quarter-floor claim, inflated monotonicity, altered anchor, replaced parent, and inflated parent-improvement census. The local bridge locks source files, certificate, manuscript, compile log, and bridge text.

These checks establish finite package integrity only. No source-uniform arithmetic L^2 estimate, uniform masked-operator theorem, fixed-power saving, Route-B reassembly, or twin-prime conclusion is obtained. The Session-named official evaluator files are absent, so the local route evaluation remains fail closed.

7 Conclusion and next question

Reciprocal centering is a simple, non-fitted scale-aware rule that improves most rows and partially repairs the $Q = 256$ block. The strongest obstruction is equally clear: 12 losses, a sub-quarter floor, and 23 nonmonotone series remain. The next minimal step is an adversarial holdout on disjoint origins and shell scales. If the repair does not transfer, the finite incidence branch should be frozen and the main route returned to the source-native arithmetic L^2 gate.

References

[1] John B. Conway. *A Course in Functional Analysis*. Springer, 2nd edition, 2000.